

Engineering a hard sol-gel adaptive coating with nonleaching antifoulant against marine biofouling in static conditions

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ABSTRACT

Covalently tethered, nonleaching antifoulant groups to surface or coating matrix are being explored as an eco-friendly option to develop a robust antifouling coating with enhanced static antifouling ability. And most of marine antifouling coatings usually have a relatively low coating hardness owing to intrinsically polymeric property. In this work, we exploited the self-assembly of random zwitterionic amphiphilic copolymers with an isothiazolinone-based antifoulant to develop a scalable hard (> 6H pencil hardness) coating against marine biofouling under static conditions. Upon exposure to aqueous media, strong fluorinate-zwitterion interactions drive microphase separation into bicontinuous networks of hydrophobic and zwitterionic nanodomains, leading to the enrichment of antifoulant moieties on the coating surface. The obtained hybrid coating could significantly restrain the growth of bacteria biofilms, the settlement of marine diatoms and mussel larvae owing to the presence of covalently tethered antifoulants. Oxidative stress levels in the body of bacteria and marine diatoms were significantly enhanced nearby the coating containing nonteaching antifoulants. On the other hand, the fouling release capability was not compromising on the developed hybrid coating with such high Young's modulus and large coating hardness, implying an obviously different antifouling mechanism. We attempt to report an ecofriendly and hard coating system to simultaneously prohibit fouling settlement and promote fouling removal in static conditions.

1. Introduction

Marine biofouling refers to unwanted attachment and accumulation of biological organisms on underwater marine equipment, and is becoming a global concern for maritime operations [1,2]. Applying antifouling coating is probably the most straightforward and cost-effective method to prohibit and control fouling organisms attachment and accumulation [3,4]. Currently, the mainstream antifouling coatings in the market are mainly involved with releasing biocides or toxic compounds into seawater, and they are gradually banned and restricted with developing marine economy and increasing ocean ecology concerns [5,6]. Silicone elastomer-based fouling release coating meets the demands of marine ecosystem conservation and realizes lasting antifouling without introducing any toxic antifoulants [3,7]. It is accepted that their good fouling release ability is arising from low surface energy and Young's modulus. The character of low surface energy generally leads to the poor substrate adhesion strength, and the relatively low hardness cannot meet the urgent demand of maritime facilities required

high impact, hydrodynamic shear and puncture resistances, such as vessel propeller and ship hull protection [8]. More seriously, fouling release capability was severely deteriorated at static conditions. In particular, soft marine biofilms developed by accumulation of bacteria and diatoms could still stick to the surface [9,10]. A slime layer was even still developed on the coated ship hull surfaces even at high navigation speeds of 30 knots [11].

To facilitate the fouling release capability in the absence of hydrodynamic shear force, amphiphilic additives [12–14], zwitterions [15–17], quaternary ammonium salts [18–21] and antifoulants [22–24] have been chemically immobilized into coating matrix. To reduce the uncontrollable antifoulant release and prolong service life, an alternative contact bioactive strategy has been developed by tethering bioactive molecules (synthetic or natural species) using covalent interaction in polymeric matrix [25–27]. However, most of antifoulant groups are readily trapped into the coating matrix, thus, cannot completely exhibit antifouling activity. And their incompatibility with polymeric matrix in seawater usually leads to the coating swelling, thus, decreasing the

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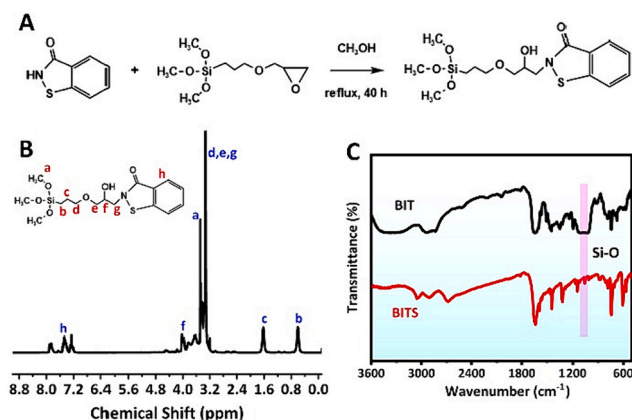


Fig. 1. A) Schematic illustration of synthesis of contact bioactive molecule 2-(2-hydroxy-3-(3-(trimethoxysilyl) propoxy) propyl) benzo[d] isothiazol-3 (2H)-one (BITS). B) ^1H NMR spectrum of BITS. C) FT-IR spectra of BIT (black) and BITS (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

mechanical property of the coating [28–32]. It is highly urgent to develop a mechanically robust coating surface enriched with covalently tethered nonteaching antifoulant groups, realizing excellent fouling release performance in static submersion conditions without strong water flow.

In this study, we firstly synthesized an ecofriendly trimethoxysilyl group-terminated antibacterial agent from a broad spectrum antibacterial agent 1, 2-benzisothiazolin-3-one (BIT). A zwitterionic amphiphilic telomer was prepared from the polymerization of 3-mercaptopropyl triethoxysilane (KH580), 2, 2, 3, 4, 4, 4-hexafluorobutyl methacrylate (HFBM) and [(methacryloyloxy) ethyl] dimethyl-(3-sulfopropyl) ammonium hydroxide (SBMA). A sol-gel hard coating was obtained by crosslinking the synthesized antibacterial agents with the zwitterionic amphiphilic telomer via condensation reaction at room temperature. Upon exposure to aqueous media, strong zwitterion-zwitterion interactions drive microphase separation into bicontinuous networks of zwitterionic and hydrophobic nanodomains, leading to the enrichment of antifoulant moieties on the coating surface. The obtained hybrid coating with up to 6H pencil hardness could simultaneous inhibit the attachment of bacteria, marine diatoms and mussel larvae, and reducing the adhesion force of settled ones. Besides, covalently tethered

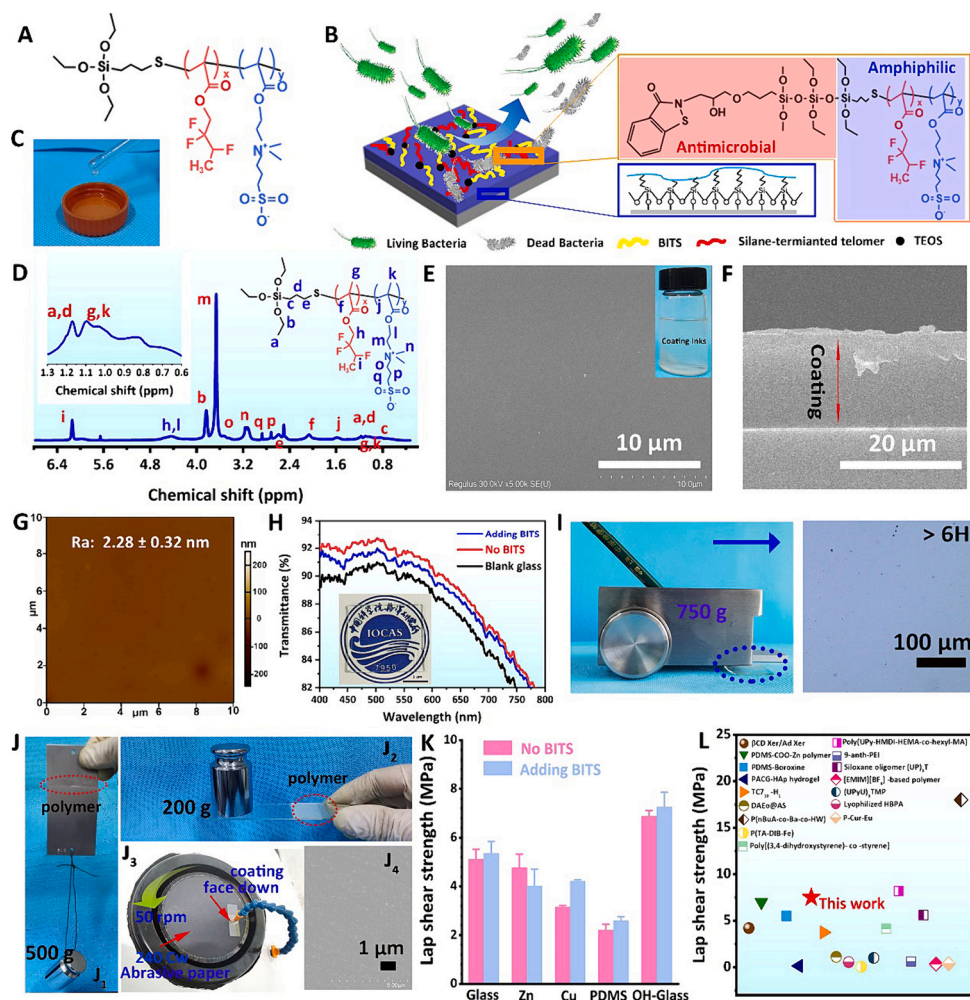


Fig. 2. A) Chemical structure and photography of the synthesized amphiphilic telomer. B) Schematic illustration of the hybrid antifouling coating. C, D) Photography (C) and ^1H NMR spectrum (D) of the amphiphilic telomer. E) SEM images of the coating and photography of the coating inks. F) Cross-section view of SEM images for the coating. G) 2D AFM topography images of the antifouling coating. H) Optical transmittance of the coatings with and without BITS. I) Pencil hardness test and corresponding microcopy photography for the antifouling coating. J) Mechanical stability test for Adding BITS coating. The coating could withstand up 500 g weight (J_1). Two glass sides are glued together using Adding BITS coating, and a 200 g weight was placed on one side of the upper glass slide (J_2). Surface abrasion test (J_3) for the coating and SEM images (J_4) of the coating after 50 abrasion cycles. K) The adhesion strength on varied substrates determined by lap-shear test. L) Comparison of the antifouling coating and previously reported adhesive materials.

nonleaching antifoulants makes the hybrid coating environmentally benign, and could not affect the metabolism and growth of non-target marine organisms. We hope this work could provide great inspirations for developing ecofriendly antifouling materials with robust antifouling ability in static conditions.

2. Results and discussion

1,2-Benzisothiazolin-3-one (BIT) and its derivatives are served as one of broad-spectrum and commercially available biocides or preservatives against microorganisms (such as bacteria, fungi, and algae) in marine antifouling and other relevant fields [33–35]. The biological activity is arising from the ability to react with intracellular sulfur-containing proteins or smaller biomolecules like glutathione, consequently leading to the death of the cells [36–38]. More importantly, these chemicals are found to have short half-life periods, making them a green and sustainable antifoulant in marine environment [39,40]. To reduce the uncontrollable release and prolong service life of antifoulants, it is necessary to immobilize bioactive molecules onto polymeric matrix. As shown in Fig. 1A, commercially available BIT molecule were firstly functionalized with trimethoxysilyl group, and a transparent light yellow liquid was obtained after reactions (Figs. 1A and S1). ^1H NMR signal from aromatic skeletal was detected, while not for N–H bond in heterocyclic ring (Figs. 1B and S2). Other characteristic peaks from (3-glycidyloxypropyl)-trimethoxysilane (KH560) were also detected for the obtained product. The molecule structure of antifoulant was further studied using FTIR spectrum. As shown in Fig. 1C, a characteristic peak at 1058 cm^{-1} assigning to stretching vibration of Si–O bonds was observed. The obtained modified antifoulant was represented as BITS.

After BIT functionalization, we try to tether trimethoxysilyl-terminated BITS with compatible polymeric antifouling matrix. We used radical polymerization to prepare an amphiphilic telomer by SBMA, HFBM and KH580. KH580 not only introduced trimethoxysilyl groups at one end of polymer chain, but also was used as chain transfer reagents and molecular weight modifier for the telomerization systems. The obtained amphiphilic telomer was represented as KH580-SBMA-HFBM, and its chemical structure and preparing route were shown in Figs. 2A and S3. As shown in Fig. 2C, the as-prepared telomer is a transparent and sticky polymer. ^1H NMR spectra of the used monomers, telomers and other polymers were also conducted (Figs. 2D and S4). As revealed by results, the signals of SBMA, KH580 and HFBM could be simultaneously detected from spectra of KH580-SBMA-HFBM telomer, indicating the successful synthesis of the telomer.

Considering the reactive trimethoxysilyl groups of telomer and BITS, we try to fabricate an amphiphilic organic–inorganic hybrid coating using silane coupling agent tetraethyl orthosilicate (TEOS), as depicted in Fig. 2B. After the telomer was mixed with BITS and TEOS, a transparent coating inks were obtained (inset in Fig. 2E). The coating inks were dropped onto a piece of glass slide and the thickness of the coating was controlled using a bar-coater, followed by curing at room temperature. The thickness of the coating on glass is $\sim 16.2\text{ }\mu\text{m}$ (Fig. 2E and F). Here, the coatings fabricated by coating inks containing BITS were represented as Adding BITS, while the coatings without adding BITS were denoted as No BITS. We investigated whether the two coatings are decomposed in marine environment by mass loss measurement after 15 day submersion in natural seawater. As shown in Fig. S5, the mass loss of each coating was less than 0.5 wt%, indicating that these two coatings are stable after long-term immersion in natural seawater, and KH580-SBMA-HFBM and BITS are successfully introduced into the hybrid networks via forming covalent bond. We further used ATR-FTIR spectra to investigate whether BITS are grafted into the hybrid networks. As shown in Fig. S6, an ATR-FTIR peak at 1635 cm^{-1} assigning to aromatic ring from BITS was observed for Adding BITS. 2D AFM topography of Adding BITS demonstrated that the average roughness (R_a) of the cured coating on glass slides was $2.28 \pm 0.32\text{ nm}$, while R_a was $2.09 \pm 0.21\text{ nm}$ for No BITS (Figs. 2G and S7). Such nanoscaled surface roughness is not only in

favor of high optical transparency, but also the ideal fouling releasing capability under shear stress. As shown in Fig. 2H, both Adding BITS and No BITS coating exhibit better optical transparency than glass slides from 400 to 800 nm wavelength. Hence, the coating could find valuable applications in underwater windows and optical instruments.

There are almost no evident scratches on Adding BITS surface after 6H pencil hardness test (Fig. 2I), which might be caused by the rigid aromatic structure of BITS and a high degree of crosslinking reaction. Actually, marine antifouling coatings with reasonable hardness could effectively resist striking and puncturing from hard fouling organisms (such as barnacles and oysters) in all directions [41]. And a relatively high hardness is desirable for marine antifouling coatings applied on ship propeller and the hard ship hull that is suitable for underwater robot cleaning [42,43], although some previous works pointed out that fouling organisms would like to land on hard surfaces other than soft ones [44,45]. In addition, adding amphiphilic telomer brought good flexibility for the final antifouling coatings because the soft segment in amphiphilic telomer could be served as a plasticizer to enable the coating precursor to form a thick but tough coating. As shown in Fig. S8, adding BITS surface exhibits a good flexibility after 100 cycles of bending test, and the coating is unbroken and still firmly adhered to substrates. However, many coating cracks could be observed on No BITS surface along the vertical direction of stress, even the coating was shed off from the substrates. To well understand the mechanisms for the good flexibility, we investigated the glass transition temperature (T_g) of KH580-SBMA-HFBM and the coating precursors. As shown in Fig. S9, A low T_g -39.9°C was detected for the amphiphilic telomer. As previously reported, a low T_g generally leads to a low crystallinity and a high movement of polymer segments. Hence, the amphiphilic telomer has low crystallinity and polymer chains could free move at room temperature. T_g of Adding BITS is 69.57°C , which is lower than that of 73.82°C for No BITS. Thereby, Adding BITS has a lower crystallinity and a greater chain mobility than No BITS, hence, exhibiting better flexibility to meet the demands of complicated marine environments. Fig. S10 shows the comparison of anti-smudge ability for Adding BITS and No BITS. An oily mark pen was used to write chirographies on the two surfaces. The writing on Adding BITS surface is clear and it is difficult to be erased by a napkin. In contrast, the writing on No BITS coating could be easily wiped off thoroughly. This is because the oily liquids could be attracted by oleophilic BITS and completely spread out on Adding BITS surface, but aggregated into small droplets on No BITS with more oleophobic zwitterion side chains.

Low surface energy fouling release coating generally are nonpolar and highly undesirable for a shipyard environment owing to low surface energy-induced nonstick property, leading to severe pollution issues in applying coating regions, coating delamination and decreased binding force of other coating species applied on a ship. Beside poor static antifouling ability, poor substrate adhesion represents another drawback to suppress the development of fouling release coatings. As shown in Fig. 2J₁, two pieces of stainless steels are tightly glued together using Adding BITS, and the formed stainless steel integration could easily lift up a weight of 500 g with an adhered area of 1 cm^2 . A 200 g weight could be placed on one side of the upper patch without any coating delamination (Fig. 2J₂). We use lap-shear test to quantitatively evaluate the adhesion strength of Adding BITS on varied substrates. As shown in Fig. 2K, a highest shear strength above 7 MPa is recorded for oxygen plasma-treated hydrophilic glass with more hydroxyl groups, because covalent bonding was formed through the condensation reaction between hydroxyl and silanol groups. In addition, the lap shear forces of coating on hydrophobic Cu and PDMS substrates were much lower than that of other substrates, owing to the absence of reactive sites for condensation. Adding BITS coatings exhibit satisfactory substrate adhesion among previously reported works (Fig. 2L), highlighting the versatility as universal adhesives [46–60]. We selected abrasion test to investigate the abrasion resistance ability of Adding BITS surface using a homemade testing platform (Fig. 2J₃). As shown in Fig. 2J₄, no obvious

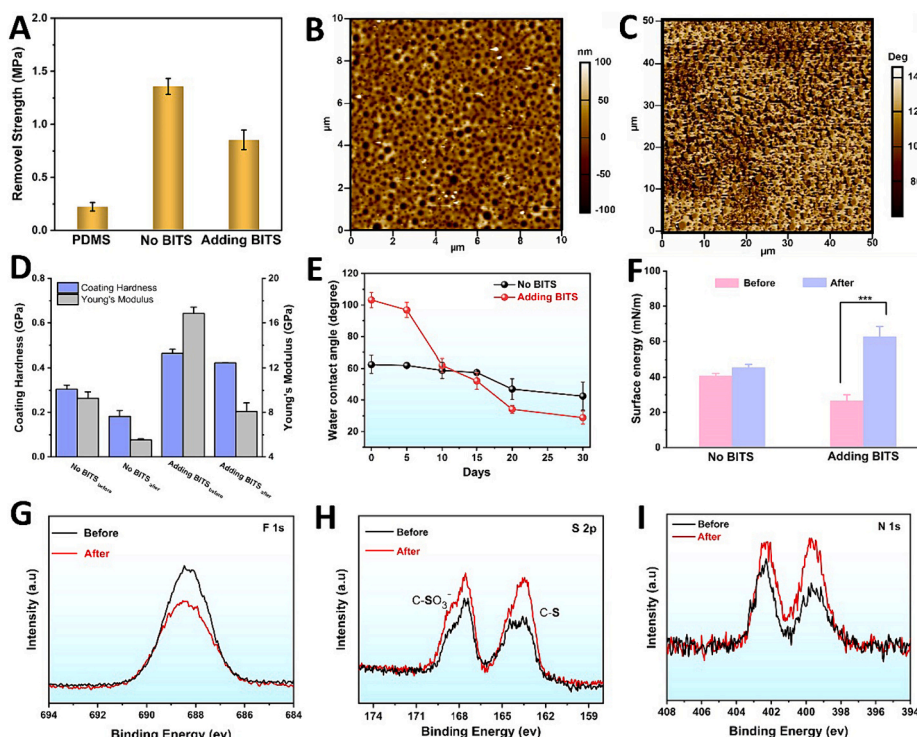


Fig. 3. A) Simulated barnacle removal strength of the coatings. B, C) AFM height and phase images of adding BITS upon exposure to aqueous media. D) Coating hardness and elastic modulus before and after 24 h water submersion. E) Water contact angle variations during immersion in natural seawater for coatings. F) Surface energy variation before and after 30 days water submersion. G–I) F 1s, S 2p and N 1s XPS spectra of Adding BITS before and after seawater immersion.

abrasion marks could be observed on hard Adding BITS surface after 50 cycles of abrasion tests with loaded weight of 100 g, 240 grit SiC sandpaper and rotating speed of 50 rpm. Otherwise, the coating preparing could be realized using spraying, dipping, and brushing coating approaches (Fig. S11).

Simulated barnacle removing experiment was conducted to assess the fouling release capability of the coatings. Five cylindrical aluminum studs with the diameter of 10 mm were glued on varied surface regions on coatings using epoxy adhesive. The adhesion strength of the epoxy adhesive is very similar with that of adhesive protein from barnacles. The shear force provided by a force gauge was horizontally applied at a rate of approximately 4.5 N s^{-1} to the adhered aluminum studs until the adhered stud edge plate detached the coating surface. The maximum shear forces were recorded in this processing and were further used to calculate the removal strength. It is accepted that a lower pseudo-barnacle removal strength on fouling release coating generally indicates a better fouling release ability [61,62]. As shown in Fig. 3A, Adding BITS exhibits a removal strength of $\sim 0.83 \text{ MPa}$, while removal strength of No BITS was $\sim 1.26 \text{ MPa}$. These results indicated that Adding BITS has better fouling release ability than No BITS. It is widely accepted that the superior fouling release capability is primarily ascribed to low elastic modulus and surface free energy [5,7]. We try to understand the detailed mechanisms for better fouling release capability of Adding BITS. As shown in Fig. 3B, the coating roughness of Adding BITS upon exposure to aqueous media is changed to $25.98 \pm 0.60 \text{ nm}$. The image exhibits a disordered bicontinuous topology with a network of zwitterionic phase (bright regions) surrounded by the hydrophobic phase (black regions). Fig. 3C shows AFM images of Adding BITS after seawater immersion, indicating that the coatings possessed a heterogeneous amphiphilic surface composed of hydrophilic and hydrophobic compositions. We studied the hardness and elastic modulus determined by nanoindentation tests. As shown in Fig. 3D, the coating hardness of Adding BITS is $\sim 0.46 \text{ GPa}$, and is decreased to $\sim 0.42 \text{ GPa}$ after immersion in natural seawater for 7 days. Coating elastic modulus is $\sim 16.80 \text{ GPa}$ for

Adding BITS, while is decreased to $\sim 9.20 \text{ GPa}$ after seawater immersion (Fig. 3C). No BITS has a coating hardness of $\sim 0.30 \text{ GPa}$, and coating hardness is decreased to $\sim 0.20 \text{ GPa}$ after seawater immersion. Elastic modulus of No BITS is $\sim 7.48 \text{ GPa}$, while decreased to $\sim 5.57 \text{ GPa}$ after seawater immersion. Obviously, both coating hardness and elastic modulus of Adding BITS are higher than those of No BITS, which might be caused by the addition of BITS molecule with rigid aromatic skeletal and the enhanced crosslinking degree. It is mentioned that the Young's modulus of Adding BITS and No BITS is on the orders of GPa, indicating that the coatings are non-elastic. Hence, Adding BITS with a higher hardness indicates a better abrasion and scratching resistance than No BITS. Fig. S12 shows the nanoindentation load-displacement curve of Adding BITS and No BITS before and after 7 days seawater immersion, which a lower depth of trace represents the higher hardness of coatings. The depth of trace left on Adding BITS is 3861 nm , but is increased to 5582 nm after 7 days seawater immersion. For comparison, the depth on No BITS is 6629 nm , while is increased to 7013 nm after water attacking. Based on the results from nanoindentation test, Adding BITS surface has higher elastic modulus and coating hardness than No BITS. Obviously, Adding BITS could have a significantly varied fouling release mechanism. In addition, both the coating hardness and elastic modulus were significantly reduced after seawater immersion-induced coating swelling.

To well understand the fouling release mechanisms of the hard coatings with high elastic modulus, wetting behavior and surface energy of the coatings were studied. As shown in Fig. 3E, water contact angles were significantly decreased from $\sim 100^\circ$ and finally reached to $\sim 28^\circ$ on Adding BITS, while were slightly decreased from $\sim 60^\circ$ and finally reached to $\sim 42.5^\circ$ on No BITS. The decreased hydrophobicity was further determined by measuring underwater oil contact angles on the coatings. As shown in Fig. S13, underwater oil contact angle on Adding BITS surface was significantly increased from $\sim 72^\circ$ to $\sim 143^\circ$, while only slight increments were observed for No BITS. The changed wetting behavior, reflected by evolutions of water and underwater oil contact

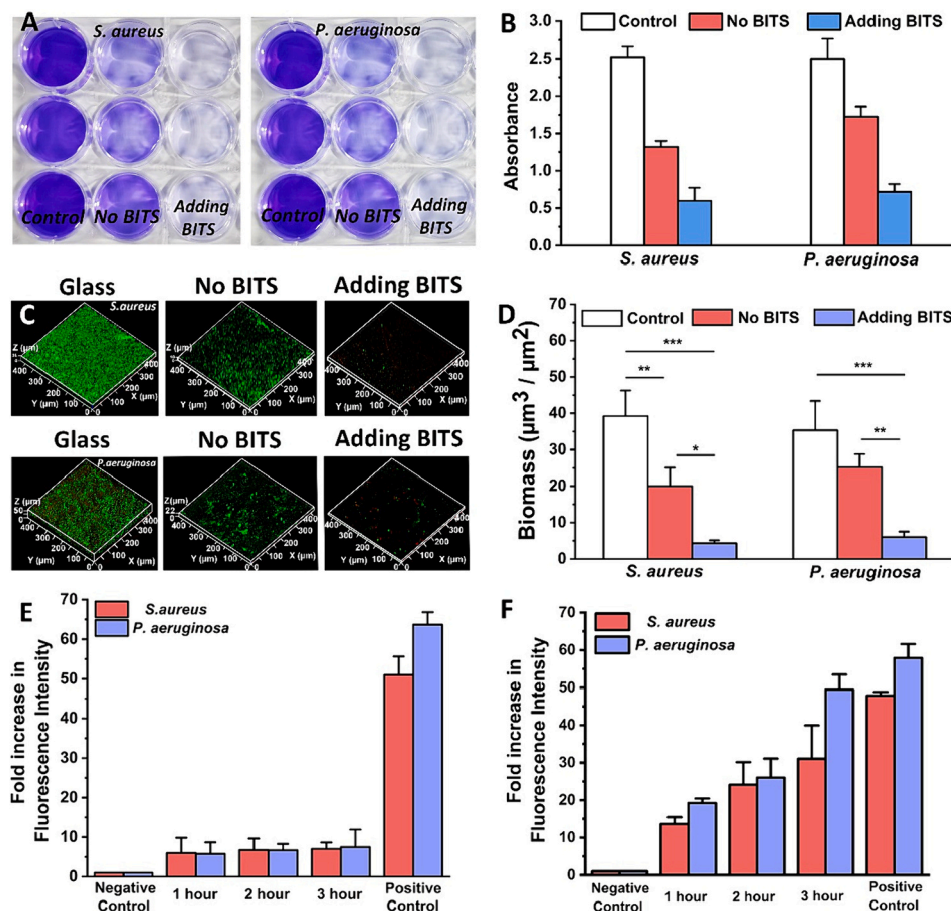


Fig. 4. A) Optical photography of *S. aureus* and *P. aeruginosa* biofilm dyed with crystal violet on different surfaces. B) The absorbance of crystal violet in *S. aureus* and *P. aeruginosa* biofilm on glass, no BITS and Adding BITS surface. C) CLSM images of STYO 9/PI live/dead assay for the attached *S. aureus* and *P. aeruginosa* bacteria on three surfaces. D) The calculated biomass of the attached bacteria on three surfaces based on CLSM images. E, F) Oxidative stress levels in two bacteria strains after incubation with No BITS (E) and Adding BITS (F) with increasing incubation time. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

angles, indicates that Adding BITS surface was turned to be hydrophilic after 30 days seawater immersion. Fig. 3F shows the evolutions of surface energy determined by calculating water and diiodomethane contact angles on the coatings. Obviously, surface energy of Adding BITS was increased, while remained nearly unchanged for No BITS. Note that the surface energy of Adding BITS was $\sim 26.3 \text{ mJ m}^{-2}$, which was very closed to the critical surface energy scope from 22 to 24 mJ m^{-2} for antifouling surfaces.

As mentioned, coating hardness, elastic modulus, surface wettability and surface energy of the nonelastic coatings were all changed after seawater immersion. Owing to the presence of KH580-SBMA-HFBM, we speculated that the changes were possibly induced by the movement of amphiphilic components with low T_g . We use ATR-FTIR and XPS spectra to investigate the water-induced surface chemistry changes on the nonelastic coating surface. As shown in Fig. S14, the signal intensity of the peak at $\sim 1060 \text{ cm}^{-1}$ assigning to Si—O bond, the peak at $\sim 1466 \text{ cm}^{-1}$ attributed to aromatic skeletal, the peak at $\sim 1600 \text{ cm}^{-1}$ coming from nitrogen heterocyclic ring of BITS and the peak at $\sim 1733 \text{ cm}^{-1}$ assigning to zwitterions were increased after seawater immersion. The signal intensity of F 1s peak at 688.2 eV was reduced on Adding BITS surface after seawater immersion (Fig. 3G). The S 2p peaks at 163.5 and 167.9 eV were assigned to C—S from heterocyclic ring of BITS and C—SO₃[−] from zwitterion segment of amphiphilic telomer, respectively. Otherwise, the signal intensity of S 2p peak was increased after seawater immersion (Fig. 3H). Similarly, the signal intensity of N 1s from the heterocyclic ring of BITS and zwitterion segment of amphiphilic telomer

were also increased after seawater immersion (Fig. 3I). Based on the results of ATR-FTIR and XPS spectra, we speculate the possible mechanisms from the viewpoint of surface chemistry for the changed coating properties. The incompatibility between the fluorinated compositions and highly polar zwitterionic segments could drive microphase separation. Driven by thermodynamics, fluorinate groups from amphiphilic telomer would gradually migrate to the air-coating surface during seawater immersion, inducing the enrichment of amphiphilic components on Adding BITS and No BITS top surface. Hydrophilic zwitterion segments would gradually come to the coating surface as low surface energy parts migrates. In this way, water contact angles on both coatings were gradually decreased, while underwater oil contact angles were gradually increased during seawater immersion. Coating hardness and elastic modulus were correspondingly declined owing to the self-enrichment of amphiphilic telomer with low T_g and crystallinity. The covalently tethering of hydrophobic BITS in hybrid antifouling coatings could also migrate to coating surface, accelerating the surface arrangement and surface-enrichment of amphiphilic telomer. Thus, Adding BITS surface could gradually become more hydrophilic and oleophobic. The more significant decline in coating hardness and elastic modulus might be caused by the enhanced surface-enrichment of amphiphilic telomer.

Adhesion of *S. aureus* and *P. aeruginosa* on different surfaces was further investigated. Fig. 4A and B shows the biofilm attachment of two bacteria strains on the coatings. The biofilm growth on No BITS and Adding BITS are significantly prohibited, and the anti-biofilm ability is

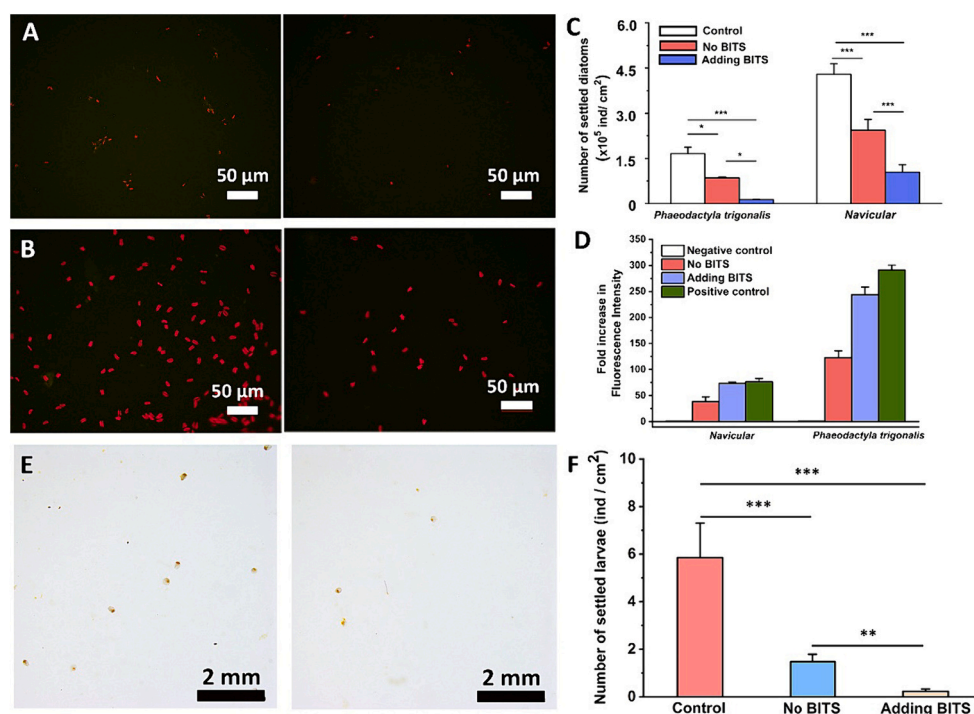


Fig. 5. A, B) Fluorescence images of adhered *Phaeodactyla trigonalis* (A) and *Navicular* (B) on No BITs (left) and Adding BITs (right) surface. C) The number of settled *Phaeodactyla trigonalis* and *Navicular* on control (PET substrate), No BITs and Adding BITs. D) The fold increase in fluorescence intensity of oxidative stress probe in two marine diatoms after incubation on varied surfaces for 1 h. E) Optical photographs of the settled *Mytilopsis sallei* larvae on No BITs (left) and Adding BITs (right). F) The number of settled larvae on control (glass slide), No BITs and Adding BITs.

evidently enhanced upon the introduction of BITs. As shown in CLSM images in Fig. 4C, only sparse and isolated single bacterial cells were observed on Adding BITs surface after 12 h exposure to suspensions of two bacteria strains. While, relatively high-density bacterial colonies were observed on No BITs surface. By quantifying the biomass on the two coating surfaces, it was found that No BITs and Adding BITs reduced the attachment of *S. aureus* by 25.16 and 90.61 %, respectively. While No BITs and Adding BITs reduced the attachment of *P. aeruginosa* attachment by 28.65 and 91.45 %, respectively (Fig. 4D). Zeta potential of the coatings was changed from ~ -23 to ~ -15 mV after introducing BITs in hard hybrid antifouling coatings (Fig. S15). We further investigated the oxidative stress-mediated antibacterial activity of No BITs and Adding BITs using an oxidant-sensing fluorescent probe 2',7'-dichlorofluorescein diacetate (DCFH-DA). The higher fluorescence intensity represents a higher level of oxidative stress in cells. As shown in Fig. 4E, the signal intensity was only slightly increased in bacteria nearby No BITs, and the fluorescence intensity was increased up to 50 folds after *P. aeruginosa* contacting Adding BITs surface for 3 h, indicating the oxidative stress levels were gradually accumulated in bacteria cells (Fig. 4F). Besides, *S. aureus* generated higher oxidative stress levels than *P. aeruginosa* because of the susceptibility to BITs.

The reduced bacteria adhesion was ascribed to the co-existence of amphiphilic telomer and BITs bioactive molecule. The amphiphilic telomer on topcoating could restrain the attachment of bacteria based on the physical barrier effect caused by the formed hydration layer. The covalently tethered BITs could significantly accelerate the oxidative stress levels in planktonic bacteria nearby Adding BITs. Oxidative stress is developed via metabolic pathways that are regarded as the resources of a reactive oxygen species (ROS) cascade. Bacteria and other microorganisms generally formed chemically reactive ROS in aerobic conditions, and ROS actually affect the bacteria physiology and biofilm forming ability. The oxidative stress response protein of microorganisms could sense the changed ROS concentration and activates the transcription of several genes involved in antioxidative defense, scavengers,

thiol redox buffers and enzymes [63,64]. When the bacteria are in contact with Adding BITs surface, they could monitor such "stimulation" induced by the enhanced oxidative stress levels caused by BITs, and preferentially escape from the surface rather than surface settlement. And the electrostatic repulsion effect between negatively charged surface and negatively charged bacteria could also reduce the bacteria attachment.

The adhesion of marine diatoms is a crucial stage in marine biofilm development and affects the settlement and colonization of macrofouling organisms. Actually, diatoms are preferentially attached and accumulated on submerged substrates that have attached proteins and bacteria in prior. *Phaeodactyla trigonalis* and *Navicular* were selected as two representative marine benthic diatoms to evaluate the anti-diatom ability of the coating. As shown in Figs. 5A–C and S16, considerable diatoms have settled and colonized on the pristine hydrophobic PET substrates, indicating that marine diatoms preferentially selected hydrophobic surface to settle and colonize. The attached cell density of *Phaeodactyla trigonalis* on PET substrate was over $1.5 \times 10^5 \text{ ind/cm}^2$, which was lower than that of *Navicular* ($\sim 4.0 \times 10^5 \text{ ind/cm}^2$) on untreated substrate. Such differences might be arising from the differences of surface property of two diatoms. The attached diatom density was largely reduced on No BITs and Adding BITs. Especially, the attached diatom density of *Phaeodactyla trigonalis* on Adding BITs was nearly reduced by an order of magnitude, compared with that on PET substrates. The settled cell density of *Phaeodactyla trigonalis* on No BITs was reached to around $1.25 \times 10^5 \text{ ind/cm}^2$. For *Navicular*, the settled cell densities on No BITs and Adding BITs were around $1.80 \times 10^5 \text{ ind/cm}^2$ and $1.1 \times 10^5 \text{ ind/cm}^2$, respectively. Oxidative stress levels in two marine diatoms were significantly accelerated when the diatoms comes into contact with Adding BITs surface in a short time (Fig. 5D). The hydration layer formed by self-enriched amphiphilic telomer on topcoating could resist diatom settlement owing to the suppressed contacting between coating and the secreted adhesive proteins. In addition, the tethered BITs in polymer network could also improve the antifouling

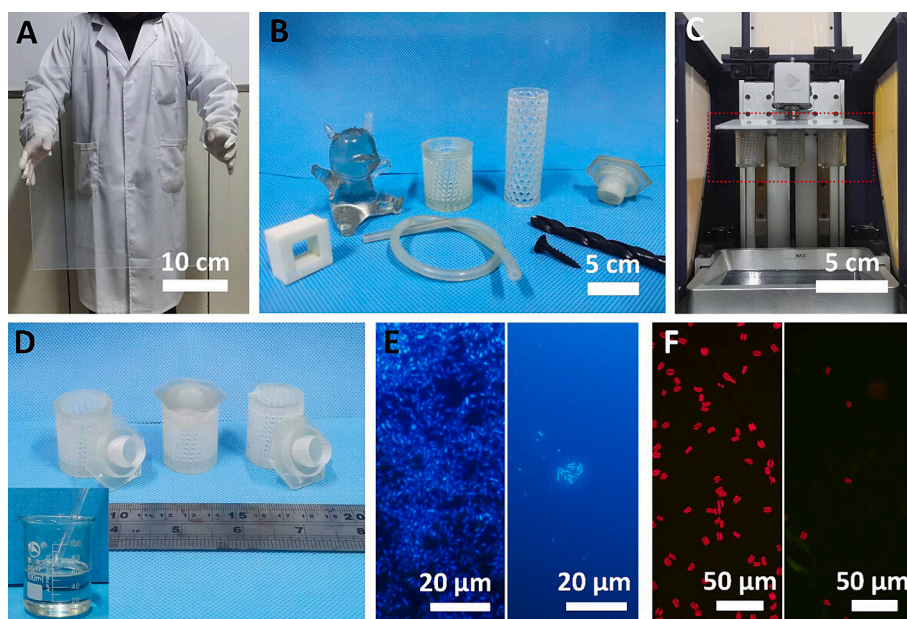


Fig. 6. A) A large antifouling coating (50 cm × 50 cm) was prepared on a large PET substrate. B) Various curved substrates with different shapes could be easily coated with the coating inks using dip coating approach. C) Photograph of the used 3D printing machine. D) The prepared 3D antifouling device and the inks used in 3D printing. E, F) Compared with pristine devices, the 3D device with antifouling inks could largely inhibit *P. aeruginosa* bacteria and *Navicular* diatom settlement.

capability under static conditions. The results of antiadhesion assays for marine diatoms correlated well with that from antibacterial assays. The undesirable settlement and colonization of mussel, barnacles and oyster on submerged marine equipment are generally regarded as “hard macrofouling”. We conducted antiadhesion assay to further evaluate the antifouling ability of our coating against marine hard macrofoulings using larvae of mussel *Mytilopsis sallei*. When the mussel was cultured for 48 h on coatings, only a few mussel settled on Adding BITS coating surface (Fig. 5E), exhibiting the highest settlement reduction of 96.10 % compared with the glass slide (Fig. 5F).

We applied Adding BITS coating on large PET substrate (50 cm × 50 cm) using brush coating method, endowing the large coating with remarkable antifouling ability after curing at room temperature (Fig. 6A). In addition, the coating can also be applied on 3D curved substrate using the simple dip coating method (Fig. 6B). The coating was intact and firmly adhered to pristine curved substrates because of the good flexibility and substrate adhesion. The coating inks exhibited good compatibility with commercial 3D printing resin, and can be constructed into 3D antifouling devices (Fig. 6C–F).

3. Conclusions

In summary, a novel antifoulant bearing trimethoxysilyl moiety, BITS, was firstly prepared via the reaction between BIT and KH560. A sol-gel hybrid coating with nonleaching antifoulant was obtained by crosslinking amphiphilic telomer of KH580, SBMA and HFBM with BITS at room temperature. The cross-linked smooth coating exhibited high transparency, and the coating hardness was up to 6H pencil hardness. The hybrid coatings exhibited ideal fouling release ability and resistance to bacterial biofilms, marine diatoms and mussel larvae. The antifoulant moieties could migrate out of the coating with the movement of fluorinate moieties, and were accumulated on coatings surface. Oxidative stress levels in the body of bacteria and marine diatoms were significantly enhanced nearby the coating surface enriched with antifoulants. When the bacteria are in contact with Adding BITS surface, they could monitor such “stimulation” induced by the enhanced oxidative stress levels, and preferentially escape from the surface. Besides, low surface energy and low surface roughness of the coating were also conducive to fouling release ability. More importantly, our coating could be applied

onto curved solid surface without comprising coating property, and they could be constructed into complex 3D devices with superior antifouling ability. The covalently tethered non-leaching antifoulants prevents its release into seawater, and makes the coating durable and ecofriendly. The excellent antifouling performance of the hard polymer-ceramic coating was caused by the self-adaptive surface chemistry, and the surface chemistry has remedied the adverse impacts from the changed coating elastic modulus and surface topographies after seawater immersion. The combination of antifoulant and amphiphilic moieties could provide powerful guidance to the preparation of hard marine antifouling coatings required high impact, hydrodynamic shear and puncture resistances, such as vessel propeller antifouling coating and underwater robot-cleaning antifouling coatings.

CRediT authorship contribution statement

Yihan Sun: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yuanyuan Shen:** Methodology. **Haibin Zhao:** Methodology. **Peng Wang:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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